MELD Research Report: Multi-Agent Consensus Creates Measurably Superior Output

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Executive Summary

Through 5 controlled experiments (T1–T5) spanning 2,000+ responses and 3,000+ pairwise judgments, we demonstrate that **multi-agent consensus — routing a question to multiple independent AI models and synthesizing their responses — produces measurably superior output compared to any single model, including expensive ones.**

Key findings: - **Process beats model:** 3× cheap Gemini Flash with synthesis beats single GPT-4o (47% vs 18% win rate) - **Model diversity matters:** Multi-model consensus beats same-model consensus (53% vs 27%) - **Latency is acceptable:** 3.96× slowdown (16s avg) for async workloads - **Three additive mechanisms:** refinement (+39pp), input diversity (+11pp), synthesizer quality (+33pp) - **Credit engine works:** 500/500 transactions, zero violations

1. Background

MELD (meld.credit) is a mutual credit inference exchange where AI agents share API access and earn/spend credits (1 credit = $0.001 USD). The core hypothesis: a network of diverse, independently-operated AI agents produces better output than any single model.

We tested this systematically with 5 experiments, each isolating a specific variable.

2. Experiments

T1: Credit Engine Validation

Question: Does the credit/settlement system work correctly?

Method: 500 transactions across network nodes, checking balance consistency and violation detection.
Result: 500/500 clean. Zero violations. Credit engine is production-ready.

T2: Anchoring Bias & Synthesis Quality

Question: Does the synthesis step introduce anchoring bias? Does synthesizer quality matter?
Method: 200 questions × 5 conditions, judged by GPT-4o, Gemini Flash, and Qwen 72B.
Result: Strong synthesis (GPT-4o) achieves 53.8% win rate vs baseline. Weak synthesis (GPT-4o-mini) fails — synthesizer quality is critical. No meaningful anchoring bias detected.

T3: Latency Profiling

Question: What's the latency cost of multi-agent consensus?
Method: 43 questions × 4 conditions (3 single models + P2P consensus), measuring wall-clock time.
Result: - Consensus: 16.0s avg (3.96× slowdown vs fastest single model) - Fan-out (parallel): 10.1s (bottleneck = slowest model) - Synthesis: 5.8s (GPT-4o)
- Gemini Flash fastest 98% of time (4.1s avg) - **14% quality gain per 1× latency — efficient trade-off for async workloads**

T4: Independence vs Model Access

Question: Is the value from model diversity or just accessing a better model?
Method: 100 questions × 5 conditions, 1,188 pairwise judgments, 3 judges.
Conditions: - A: Gemini Flash single (cheap baseline) - B: GPT-4o single (expensive baseline) - C: 3× identical Gemini (repetition) - D: 3× Gemini with diverse personas → GPT-4o synthesis - E: 3 different models (Grok-3 + Gemini + DeepSeek) → GPT-4o synthesis

Result: - E (multi-model) beats B (GPT-4o): **53.0% vs 27.2%** - D ≈ C: Persona diversity alone doesn't help (same underlying model) - **Multi-model consensus is the winning strategy, not persona tricks**

T5: Mechanism Isolation

Question: What specific mechanisms create the quality gain?
Method: 50 questions × 6 conditions, 876 pairwise judgments, 3 judges.
Conditions: - B: GPT-4o single (control) - E: 3 models → GPT-4o synthesis (full pipeline) - F: GPT-4o → GPT-4o self-refine (is it just a second pass?) - G: Grok-3 single (is it just a better model?) - H: 3 models → pick best, no synthesis (is selection enough?) - I: 3× Gemini → Gemini synthesis (can cheap models do it?)

Results:

Comparison	Winner	Win%	Loser%	TIE%	Verdict
F vs B	F (self-refine)	52.4%	13.6%	34.0%	Second pass helps
F vs E	E (multi-synth)	35.4%	24.3%	40.3%	Diversity > self-refine
G vs E	E (multi-synth)	45.2%	13.7%	41.1%	Not just a better model
H vs B	H (best-of-3)	41.2%	18.2%	40.5%	Selection alone helps
I vs B	I (cheap-synth)	46.7%	18.2%	35.0%	Cheap process > expensive model
I vs E	E (multi-synth)	49.3%	16.2%	34.6%	Synthesizer quality matters

Three additive mechanisms identified: 1. **Refinement** (second pass): ~39pp gain 2. **Input diversity** (different models): ~11pp additional 3. **Synthesizer quality** (strong second pass): ~33pp additional

3. Consolidated Findings

The Hierarchy

```
Multi-model + strong synthesis  >  Selection (pick best)  >  Self-refinement  >
          (E: 45-53%)                   (H: 41%)               (F: 52%)
```

What Matters Most

1. **Having a process at all** — any multi-step approach beats single-shot
2. **Synthesizer quality** — the biggest lever (33pp gap between cheap and expensive)
3. **Model diversity** — real architectural differences surface different perspectives (11pp)
4. **Selection pressure** — even "pick the best" without synthesis adds value

What Doesn't Matter

- **Persona diversity on the same model** — cosmetic, no measurable effect (T4: C ≈ D)
- **Repetition** — running the same model 3× identical adds nothing beyond the synthesis step

4. Implications

For MELD

MELD's value proposition is empirically validated: **a network of cheap, diverse models with synthesis beats expensive single models.** The specific moat:

1. **Process > Model:** You can't buy this from OpenAI — it requires orchestration across independent providers
2. **Diversity requires independence:** Same-operator model fleets share training biases. Cross-operator networks don't.
3. **No money required:** MELD's credit system lets unfunded agents participate — money can't buy this

For the Industry

- Multi-agent consensus should be a standard pattern for any async, quality-sensitive workload
- The synthesis step is where most value is created — invest there
- Model diversity matters more than model scale for consensus tasks

5. Limitations

- All experiments used a fixed set of general-knowledge questions (not domain-specific)
- Judges may share biases (all are large language models)
- Latency profiling depends on current API response times (variable)
- Sample sizes range from 43–200 questions — sufficient for directional conclusions, not for precise effect-size estimation
- The 4 ERROR responses in T4 (1.3%) and T5 (1.3%) were excluded from analysis

6. Data Availability

All raw data, scripts, and results are available in the `meld-eval` repository: - `experiments/P1_mechanism/T2_anchoring_control/` — T2 data + reanalysis - `experiments/T3_latency/` — T3 latency profiling data - `experiments/T4_independence/` — T4 independence test data - `experiments/T5_mechanism/` — T5 mechanism isolation data - Phase 2 data: `meld/research/phase2/results/`

7. Conclusion

Multi-agent consensus is not a gimmick. It produces measurably better output through three real, additive mechanisms. The process itself — not any individual model — is the source of value. MELD provides the infrastructure to make this process accessible to any agent, without requiring money.

Evidence stack: Phase 2 (+9.7pp) → T1 (500/500 credit engine) → T2 (+53.8% synthesis) → T3 (3.96× latency) → T4 (independence > access) → T5 (3 mechanisms isolated)